John Paul R. Martirez

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TECHNICAL & RESEARCH PROJECTS

FORENSPECT | Digital Image Forensics Framework **January 2026 – Present**
Thesis Project

- Developed a dual-branch deep learning framework using PyTorch, EfficientNet-B0, and TruFor-inspired Transformers to detect AI-generated and manipulated images.
- Built a cross-platform forensic application using FastAPI, Next.js 16, and Tauri v2, with automated PDF report generation and visual forensic overlays.

Ephnyr AI | Full-Stack Multi-Tenant RAG Platform **August 2026**
AI & System Architecture

- Built a full-stack multi-tenant RAG platform using Next.js, FastAPI, LangChain, PostgreSQL, pgvector, and 384-dimensional FastEmbed embeddings for document search and retrieval.
- Integrated Groq LLM for fast AI responses with document citations, separate knowledge rooms, temporary session data, and automated storage quotas.

FocusFolio | AI Study Workspace & Document Quiz Generator **July 2026**
EdTech & AI Platform

- Built a system using Next.js, FastAPI, LangChain, Pinecone, and Google Gemini that automatically converts uploaded PDF lessons into interactive quizzes and flashcards.
- Created a split-screen study workspace with a built-in PDF reader and an AI tutor that strictly answers questions based on the uploaded document to keep students focused.

CheckMatePH | Political Fact-Checking & Civic Engagement Platform **May 2026**
SIKAPTala 2026 Hackathon | Top 10 Finalist (7th Place) | De la Salle University - Dasmariñas

- Developed an AI-driven fact-checking pipeline that evaluates submitted claims against verified public records to combat online misinformation for Filipino voters.

EDUCATION

Manuel S. Enverga University Foundation | Lucena City, Quezon
Bachelor of Science in Computer Science

- **Relevant Courses:** Data Structures & Algorithms, Object-Oriented Programming, Programming Languages, Software Engineering, Database Management Systems, Software Testing, Cloud Computing, Machine Learning, Intelligent Systems
- **Honors & Scholarships:** DOST-SEI Scholar, University Scholar, Consistent Dean's Lister
- **Academic Standing:** Cumulative GPA: 1.197 | Expected Graduation: June 2027

LEADERSHIP EXPERIENCE

MSEUF Association of Multimedia Artists and Technologists
4th Year Representative | BSCS **July 2026 – Present**
3rd Year Representative | BSCS **July 2025 – May 2026**

- Acted as the primary representative between CS students and executive officers, communicating academic concerns and coordinating department-wide technical activities.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL
Frameworks & Libraries: React, Next.js, FastAPI, Node.js, Tailwind CSS, LangChain
Databases & Backend: Supabase, PostgreSQL, RESTful APIs
Cloud & Deployment: Microsoft Azure Fundamentals, Vercel
Developer Tools: Git, GitHub, Postman, CI/CD basics
Testing & Quality Assurance: Black-box Testing, White-box Testing, User Acceptance Testing (UAT)

CERTIFICATIONS & TRAINING

- Generative AI: Introduction and Applications — IBM, Coursera (2026)
- Machine Learning with Python — IBM, Coursera (2026)
- Working with the OpenAI API — DataCamp (2026)
- Prompt Engineering with the OpenAI API — DataCamp (2026)
- The Complete Full-Stack Web Development Bootcamp (61.5 hrs) — Udemy (2025)
- ASEAN Data Science Explorers — SAP & ASEAN Foundation (2026)
- iSITE AI Hackathon — HEXCOREPH Labs & iSITE (2026)
- SiKAPTala 2026 Virtual Hackathon — DLSU-D (2026)
- National Programming Challenge — CodeChum (2025)
- Open iT Codefest — OpeniT Philippines (2025)
- Tech Nexus 2024 — Campus DEVCON (2024)